

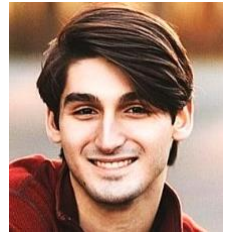
KEVIN RIVERA

ORLANDO, FLORIDA

(863) 722 6776

KEVINANDRES2069@GMAIL.COM

LINKEDIN.COM/IN/KEVINRIVERA1975



EDUCATION

DUKE UNIVERSITY

August 2026 - Present

Masters of Engineering in Robotics & Autonomy

UNIVERSITY OF CENTRAL FLORIDA

May 2022 - May 2026

Bachelor of Science, Mechanical Engineering

GPA: 3.51

Campus Involvement: ASME, ASHRAE, BMES, IISE, SHPE, Material Advantage

EXPERIENCE

AGRICULTURAL ROBOTICS LAB – UCF

May 2025 – August 2026

Volunteer Undergraduate Research Assistant

- Contributed to development of a novel, low-cost final-approach depth estimation system for a strawberry-harvesting robot using a decoupled dual RGB camera setup, deterministic color-channel image processing, and a polynomial estimation model calibrated via SSE minimization
- Achieved sub-millimeter depth estimation accuracy (0.443 mm RMS, ~1% normalized error) across a 2.5–6.5 cm target range without RGB-D cameras, LiDAR, or neural networks
- Designed and implemented the full real-time image processing pipeline in MATLAB including weighted RGB channel decomposition, grayscale conversion, binary thresholding, morphological hole-filling, noise removal, kernel smoothing, and perimeter extraction for fruit geometry detection; collaborated on remaining vision and estimation system components
- Designed all SolidWorks models for two 3D-printed test fixtures and end-effector integration using a modular architecture, allowing individual components to be replaced or modified independently to minimize reprinting costs
- Collaborated on Raspberry Pi integration for motor control and camera interfacing across a two-camera RGB system
- Assigned as one of two initial members on the research effort, later expanding to a three-person team; delivered weekly progress reports and technical recommendations to lab director
- Co-authoring a research paper with the lab for journal publication and/or conference presentation
- Conducted ~10 hours/week of hands-on lab work including hardware testing, safety checks, component replacement, and iterative debugging of vision and estimation systems

PROJECT – WIZARDRY CHESS

January 2026 – May 2026

- Wizardry Chess is a fully automated physical chess board that synchronizes with live online games and moves pieces in real time. Built using a custom XY gantry system, stepper motors, and an electromagnet housed beneath a laser-cut wooden board, the system physically replicates moves from online matches played through the Lichess API.

- The board is powered by a Raspberry Pi, which manages API communication, real-time game state tracking, and motor control. When a move is made online, the system translates chess notation into precise X-Y coordinates and actuates the gantry to move the correct piece across the board.
- What makes Wizardry Chess unique is its “Loyalty Mechanic.” Using live engine analysis from Lichess, the system evaluates player performance. If a player makes too many blunders, their pieces may become “insubordinate” and execute a randomly selected legal move instead. A visual loyalty indicator keeps players informed of their standing, adding a strategic and interactive twist to traditional chess.

Key Features:

- Real-time online game synchronization
- Custom-built gantry and electromagnet actuation system
- Raspberry Pi-based embedded control system
- API integration and move parsing
- ROS2
- Experimental game mechanics (Loyalty System)

This project combines mechanical design, embedded systems, API integration, and real-time control systems into a fully functional, interactive chess experience.

UCF SENIOR DESIGN PROJECT - BRING ROAR TO LIFE

August 2025 – Present

Anatomically correct skeleton: animatronic Stegosaurus, from concept to fully implemented prototype

- Technical memoranda on digital and control systems re: control theory with theoretical analysis, hardware (microcontrollers, microcomputers, and microprocessors), robotics programming and operation systems (algorithmic and mathematical methods – controlled/autonomous robotics systems)
- Engineering design specifications (Customer Requirements, Competitor Benchmarking, Design Requirements, Functional Decomposition, Component Decomposition, House of Quality)
- Contacted Lawrence Hall of Science, Dr. Matt Dempsey, and Dr. Susannah Maidment (London Natural History Museum) for skeletal design validation and joint movements
- Meetings with Lawrence Hall of Science (Berkeley, CA) re: donating the project for educational purposes
- Monday meetings with adjunct professor from NASA
- Tuesday meetings (5-person team) re: logistics
- 150-page design report before implementation and modelling
- Customer vs design requirements matrix; “House of Quality”
- Requirements decomposition, functional decomposition, component decomposition, Gantt chart
- Design cost estimations (min. and max.), incl. material costs, power systems, control systems, hardware components
- Risk assessment with museum; stakeholder and user research; competitor comparison
- Developed/wrote the code for the robot
- Weekly progress reports
- Operating and Maintenance manuals for museum staff

Product Engineer

- Developed and contributed to a company patent for innovative roadway lighting solutions decreasing cost while improving safety and energy efficiency.

NON-PROFIT - PHI ETA MU

August 2024 - Present

President (All U.S. Chapters)

- Ran “directive” meetings: profit and participation reports; amendments and annulments of fraternity Reglamiento; review/signoff: accounting books, records, reporting documents
- Operations, administration (Delta Chapter)
- Launched merchandise store for sustainable revenue stream; designed new merchandise, managed pricing, costs, and profits
- New, collaborative cross-chapter social event (Universal’s Halloween Horror Nights)
- Membership/reunion events (Miami, Tampa, Bronson, Lakeland, Orlando, Iowa, Texas, Puerto Rico)
- Year-end camping trip (60 attendees)
- Alumni event (25 attendees)
- Sunday membership meetings
- Monthly reports to Higher Council

Secretary

- Organizational records, meetings, chapter communications
- Event/Initiative coordination (expanded chapter engagement)
- Weekly logistics meetings
- Monthly reports to the council
- Designed posters for every event

VINCI CONSTRUCTION – HUBBARD

May 2024 - January 2025

Intern

- Cost analyses, estimates for transportation, permitting, and implementation of asphalt projects in private and governmental jobs as primary contractor and/or subcontractor.
- Clients: government and private sector (Universal Studios, Walt Disney World, FDOT, Orlando International Airport, Orlando Sanford International Airport, Orange County, Seminole County, Osceola County, Brevard County, Polk County)
- Weekly reports, department review meetings

DR. SOHN MATERIALS SCIENCE AND ENGINEERING RESEARCH GROUP (UCF)

Undergraduate Research Volunteer

November 2022 - May 2024

- Materials development: preparation, testing of 3D-printed samples
- Code Development: AI tool for predicting material properties
- Data analysis: optimal printing parameters
- Advanced lab operations (electron microscopes, SLM printer, gas atomization)

SKILLS

- **SOFTWARE:** MATLAB, Python, C++, AutoCAD, SolidWorks, Fusion360, HCSS HeavyBid, BlueBeam, Photoshop, Canva, Inkscape, Microsoft Suit, ImageJ, Fiji,

- **LAB INSTRUMENTATION:** Laser melting printers, gas atomization, metal polishing, SEM, XRD, Scientific Furnace, Arduino, Rasbery Pi (Pico), ESP32
- **LANGUAGES:** Spanish and English